

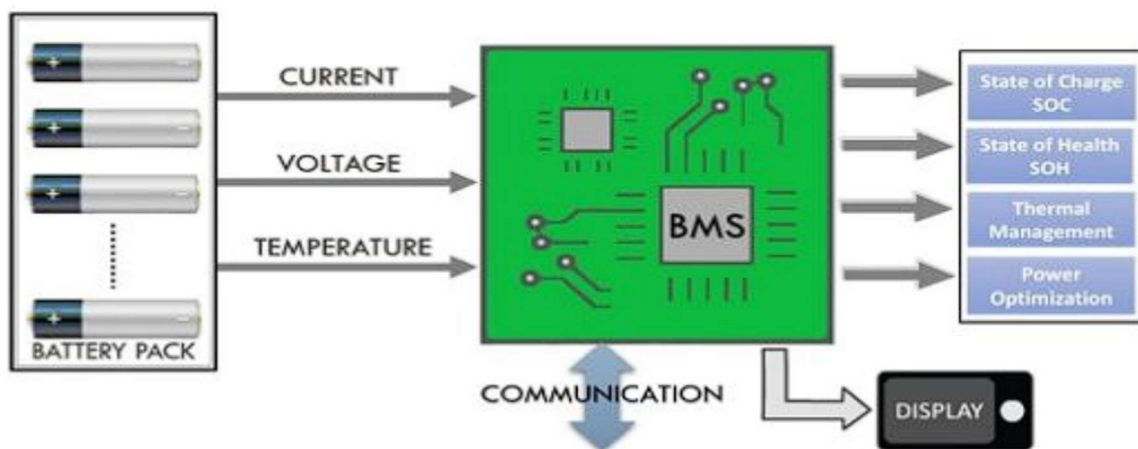
CHAPTER 1: INTRODUCTION

1.1 Background

With the rapid growth of portable electronics, electric vehicles, and renewable energy systems, batteries have become an essential component of modern technology. Among various types, lithium-ion batteries are widely used due to their high energy density, lightweight nature, and long-life cycle. However, these batteries are highly sensitive to operating conditions such as temperature, voltage, and current.

Improper handling of batteries can lead to serious issues like overheating, overcharging, deep discharge, and even thermal runaway, which may result in fire hazards. To ensure safe and efficient operation, a Battery Management System (BMS) is required. A BMS continuously monitors battery parameters and protects the system by controlling charging and discharging processes.

This project focuses on designing a simple and cost-effective BMS using an NTC thermistor for temperature sensing, a shunt resistor for current measurement, and a MOSFET for switching and protection.



Battery Management System

1.2 Problem Statement

Lithium-ion batteries, despite their advantages, are prone to safety risks if not properly monitored. Overcharging, excessive current flow, and high temperature conditions can degrade battery performance and significantly reduce its lifespan. In extreme cases, these conditions may lead to hazardous situations such as battery swelling, leakage, or explosion.

Commercial Battery Management Systems are often complex and expensive, making them unsuitable for small-scale or low-cost applications. Therefore, there is a need to develop a simple, reliable, and economical BMS that can effectively monitor key parameters such as temperature and current, and provide protection against unsafe operating conditions.

1.3 Social & Environmental Relevance

The development of a Battery Management System has significant social and environmental importance. With the increasing use of batteries in electric vehicles, solar energy storage systems, and portable electronics, ensuring battery safety is critical for both users and the environment.

From a social perspective, a reliable BMS helps prevent accidents caused by battery failures, thereby enhancing user safety. It also improves the reliability of devices that people depend on in daily life.

Environmentally, efficient battery management increases battery lifespan, reducing the frequency of battery disposal. This helps in minimizing electronic waste and lowering the environmental impact associated with battery production and disposal. Additionally, safer and longer-lasting batteries support the adoption of renewable energy systems, contributing to sustainable development.

1.4 Scope

The scope of this project is to design and implement a basic Battery Management System capable of monitoring essential battery parameters and providing protection against unsafe conditions.

The system focuses on:

- Temperature monitoring using an NTC thermistor
- Current sensing using a shunt resistor
- Voltage monitoring using Arduino
- State of Charge (SOC) and State of Health (SOH) estimation
- Cell balancing for improved battery performance
- Display of battery parameters using an LCD
- Automatic cutoff using a MOSFET under abnormal conditions

This project is implemented using an Arduino-based control system, enabling real-time monitoring and decision-making. It includes advanced features such as SOC and SOH estimation, cell balancing, and user-friendly display through an LCD module. The proposed system is suitable for small-scale applications such as portable devices, rc cars, power banks, and basic energy storage systems.

CHAPTER 2: LITERATURE SURVEY

2.1 Literature Review

Battery Management Systems (BMS) have been widely researched and developed to ensure the safe and efficient operation of batteries, especially lithium-ion batteries. Various approaches have been proposed in literature, ranging from simple analog protection circuits to advanced microcontroller-based intelligent systems. Early BMS designs focused primarily on basic protection mechanisms such as overvoltage, undervoltage, and overcurrent protection using discrete components like comparators and relays. These systems were simple and cost-effective but lacked accuracy and flexibility.

With technological advancements, modern BMS designs incorporate microcontrollers and embedded systems to enable real-time monitoring and control. These systems use sensors to measure parameters such as voltage, current, and temperature. Advanced algorithms are implemented to estimate State of Charge (SOC) and State of Health (SOH), improving battery performance and lifespan.

Temperature monitoring using NTC thermistors is a commonly adopted technique due to its simplicity and reliability. Similarly, shunt resistors are widely used for current sensing because of their accuracy and low cost. MOSFETs are preferred for switching and protection due to their fast response and high efficiency.

Recent developments also include smart BMS with wireless communication, data logging, and integration with IoT platforms. However, these systems are often complex and expensive, making them less suitable for small-scale applications.

2.2 Limitations of Existing Methods

Despite significant advancements in BMS technology, several limitations still exist in current systems.

Firstly, many advanced BMS solutions are expensive and require complex hardware and software, making them unsuitable for low-cost or educational applications. This creates a gap for simple and economical systems that can still provide essential monitoring and protection features.

Secondly, basic analog systems lack flexibility and cannot adapt to varying conditions. They also do not provide real-time data or advanced functionalities such as SOC and SOH estimation.

Thirdly, while smart BMS systems offer advanced capabilities, they often consume more power and require additional infrastructure such as communication modules and cloud platforms.

Another limitation is that many systems focus either on voltage or current monitoring, while integrated monitoring of temperature, current, and voltage is not always implemented in low-cost designs.

The proposed project aims to address these gaps by developing a cost-effective and efficient BMS that integrates temperature sensing, current measurement, voltage monitoring, and protection mechanisms using readily available components and an Arduino-based control system.

CHAPTER 3: SYSTEM DESIGN & METHODOLOGY

3.1 Proposed Methodology

The proposed Battery Management System (BMS) is designed to monitor and control critical battery parameters such as voltage, current, and temperature in real-time. The system integrates sensing, processing, and control units to ensure safe battery operation.

Block Diagram Description

The overall system can be represented using the following functional blocks:

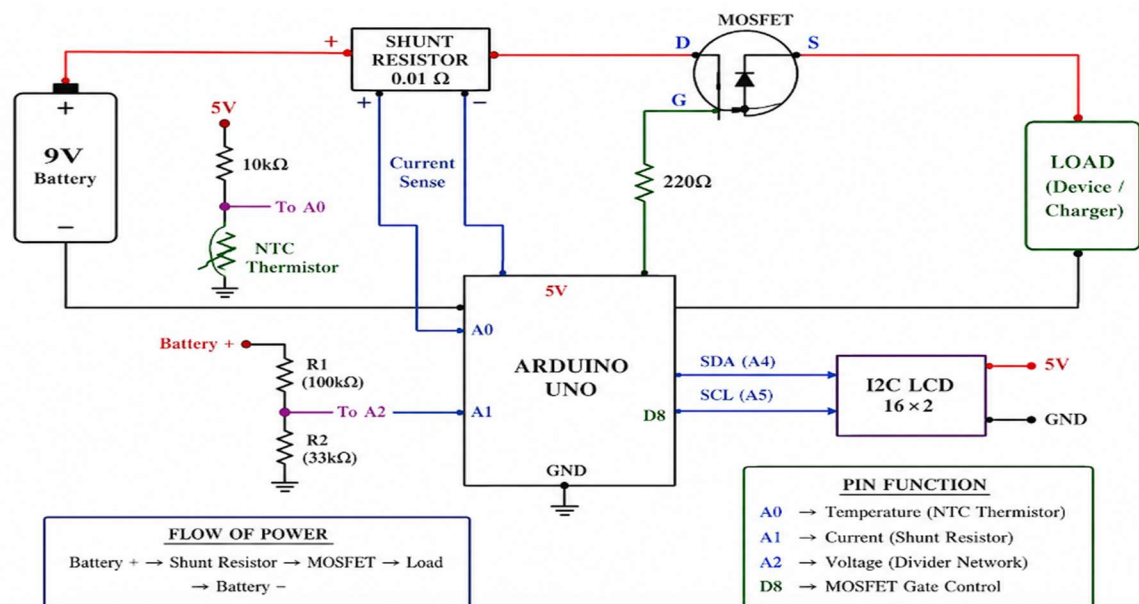
Battery → Sensors → Arduino Controller → Decision Logic → MOSFET Switch → Load

- **Battery:** Provides power to the system and is the main element being monitored.
- **Sensors:**
 - NTC Thermistor for temperature sensing
 - Shunt Resistor for current measurement
 - Voltage divider circuit for voltage sensing
- **Arduino Controller:** Acts as the brain of the system, collecting data from sensors and processing it.
- **Decision Logic:** Predefined threshold values are used to determine safe operating conditions.
- **MOSFET Switch:** Disconnects the battery from the load during unsafe conditions.

- **Display (LCD):** Shows real-time values of voltage, current, temperature, SOC, and SOH.

Flowchart Description

1. Start the system
2. Initialize Arduino and sensors
3. Read battery parameters (Voltage, Current, Temperature)
4. Calculate SOC and SOH
5. Display values on LCD
6. Compare values with predefined thresholds
7. If parameters are within safe limits → Continue operation
8. If any parameter exceeds safe limits → Turn OFF MOSFET (cutoff)
9. Repeat monitoring continuously



Flow of our project

This methodology ensures continuous monitoring and immediate response to unsafe conditions, thereby protecting the battery.

3.2 Tool and Technique Selection

The selection of tools and techniques in this project is based on simplicity, cost-effectiveness, availability, and reliability.

1. Arduino (Microcontroller)

Arduino is chosen as the main controller due to its ease of programming, open-source nature, and wide community support. It allows real-time data acquisition and control without requiring complex development environments.

2. NTC Thermistor

An NTC (Negative Temperature Coefficient) thermistor is used for temperature sensing because its resistance decreases with an increase in temperature. It is cost-effective, highly sensitive, and suitable for battery temperature monitoring.

3. Shunt Resistor

A low-value shunt resistor is used for current sensing. It provides accurate current measurement by producing a small voltage drop proportional to the current flowing through it. It is simple and widely used in practical systems.

4. MOSFET

A MOSFET is used as a switching device due to its high efficiency, fast switching speed, and low power loss. It enables automatic cutoff of the battery during unsafe conditions such as overcurrent or overheating.

5. Voltage Divider Circuit

A voltage divider is used to scale down battery voltage to a level that can be safely measured by the Arduino analog input pins.

6. LCD Display (I2C)

An I2C-based LCD is used for displaying real-time battery parameters. It reduces wiring complexity and improves system readability.

Justification of Overall Approach

The combination of these components results in a system that is:

- Low-cost and easy to implement
- Reliable for basic battery monitoring
- Scalable for future improvements
- Suitable for educational and small-scale applications

This methodology ensures a balance between performance and simplicity while addressing the key requirements of a Battery Management System.

CHAPTER 4: IMPLEMENTATION & TESTING

4.1 Experimental Setup

The Battery Management System (BMS) was implemented using a combination of sensing components, control unit, and output display. The setup was designed to monitor battery parameters in real-time and provide protection against unsafe operating conditions.

Hardware Components Used

- Arduino Uno (Microcontroller)
- NTC Thermistor (Temperature sensing)
- Shunt Resistor (Current sensing)
- MOSFET (Switching and protection)
- Lithium-ion Battery
- Voltage Divider Circuit (for voltage sensing)
- I2C LCD Display (16x2)
- Connecting wires, breadboard, and power supply

Circuit Description

The battery is connected to the sensing circuit where:

- The **NTC thermistor** is placed in contact with the battery surface to measure temperature.
- The **shunt resistor** is connected in series with the load to measure current.

- A **voltage divider circuit** is used to step down the battery voltage to a measurable level for the Arduino.

All sensor outputs are fed into the Arduino's analog input pins. The Arduino processes these signals and calculates voltage, current, temperature, State of Charge (SOC), and State of Health (SOH).

Based on predefined threshold values:

- If parameters are within safe limits → MOSFET remains ON
- If any parameter exceeds safe limits → MOSFET turns OFF, disconnecting the load

The **LCD display** shows real-time values for monitoring purposes.

Software Implementation

The Arduino is programmed using embedded C (Arduino IDE). The code performs:

- Analog data acquisition from sensors
- Conversion of sensor values into real-world units
- Threshold comparison
- Control of MOSFET switching
- Display of parameters on LCD

4.2 Output

Test Case 1: Normal Operating Condition

- **Input:** Normal temperature, voltage, and current
- **Expected Output:** MOSFET ON, system operates normally

- **Observed Result:** System displayed correct values and operated without interruption

Test Case 2: High Temperature Condition

- **Input:** Increased temperature (simulated using heat source)
- **Expected Output:** MOSFET OFF (cutoff triggered)
- **Observed Result:** System successfully detected temperature rise and disconnected the load

Test Case 3: Overcurrent Condition

- **Input:** Increased load causing high current
- **Expected Output:** MOSFET OFF
- **Observed Result:** Current threshold exceeded and system turned OFF the circuit

Test Case 4: Overvoltage Condition

- **Input:** Voltage beyond safe limit
- **Expected Output:** System protection activated
- **Observed Result:** Voltage was detected correctly and MOSFET cutoff was triggered

Arduino Code

```
#include <Wire.h>
```

```
#include <LiquidCrystal_I2C.h>
```

```
#include <math.h>
```

```

// ===== LCD SETUP =====

LiquidCrystal_I2C lcd(0x27, 16, 2);

// ===== PIN DEFINITIONS =====

#define CELL1_PIN A0

#define CELL2_PIN A1

#define TEMP_PIN A2

#define MOSFET_PIN 7

// ===== CONSTANTS =====

#define VREF 5.0

#define ADC_RES 1023.0

// Voltage divider values

#define R1 100000.0

#define R2 10000.0

// Thermistor constants

#define SERIES_RESISTOR 10000.0

#define NOMINAL_RESISTANCE 10000.0

#define NOMINAL_TEMPERATURE 25.0

#define B_COEFFICIENT 3950.0

```

```

// Safety limits

#define MAX_CELL_VOLT 4.2

#define MIN_CELL_VOLT 3.0

#define MAX_TEMP 60.0

// ===== FUNCTIONS =====

// Read voltage with averaging (stable readings)

float readVoltage(int pin) {

    long sum = 0;

    for (int i = 0; i < 10; i++) {

        sum += analogRead(pin);

        delay(2);

    }

    float adc = sum / 10.0;

    float v = (adc * VREF) / ADC_RES;

    return v * ((R1 + R2) / R2);

}

// Read temperature (NTC)

float readTemperature() {

    int adc = analogRead(TEMP_PIN);

    float voltage = (adc * VREF) / ADC_RES;

```

```

float resistance = SERIES_RESISTOR * ((VREF / voltage) - 1);

float steinhart;

steinhart = resistance / NOMINAL_RESISTANCE;

steinhart = log(steinhart);

steinhart /= B_COEFFICIENT;

steinhart += 1.0 / (NOMINAL_TEMPERATURE + 273.15);

steinhart = 1.0 / steinhart;

steinhart -= 273.15;

return steinhart;

}

// ===== SETUP =====

void setup() {

  Serial.begin(9600);

  pinMode(MOSFET_PIN, OUTPUT);

  digitalWrite(MOSFET_PIN, HIGH); // MOSFET ON initially

  // I2C init

  Wire.begin();

  lcd.init();

  lcd.backlight();

```

```

lcd.setCursor(0, 0);

lcd.print("BMS Initializing");

delay(2000);

lcd.clear();

}

// ===== LOOP =====

void loop() {

    // Read sensors

    float cell1 = readVoltage(CELL1_PIN);

    float packVoltage = readVoltage(CELL2_PIN);

    float cell2 = packVoltage - cell1;

    float temperature = readTemperature();

    bool fault = false;

    // ===== PROTECTION =====

    if (cell1 > MAX_CELL_VOLT || cell2 > MAX_CELL_VOLT) {

        fault = true;

        Serial.println("Over Voltage!");

    }

    if (cell1 < MIN_CELL_VOLT || cell2 < MIN_CELL_VOLT) {

        fault = true;

```

```

    Serial.println("Under Voltage!");
}

if (temperature > MAX_TEMP) {

    fault = true;

    Serial.println("Over Temperature!");

}

// MOSFET control

digitalWrite(MOSFET_PIN, fault ? LOW : HIGH);

// ===== SERIAL OUTPUT =====

Serial.print("C1: "); Serial.print(cell1, 2);

Serial.print(" V | C2: "); Serial.print(cell2, 2);

Serial.print(" V | Temp: "); Serial.print(temperature, 1);

Serial.println(" C");

// ===== LCD DISPLAY (NO FLICKER) =====

lcd.setCursor(0, 0);

lcd.print("C1:");

lcd.print(cell1, 2);

lcd.print(" C2:");

lcd.print(cell2, 2);

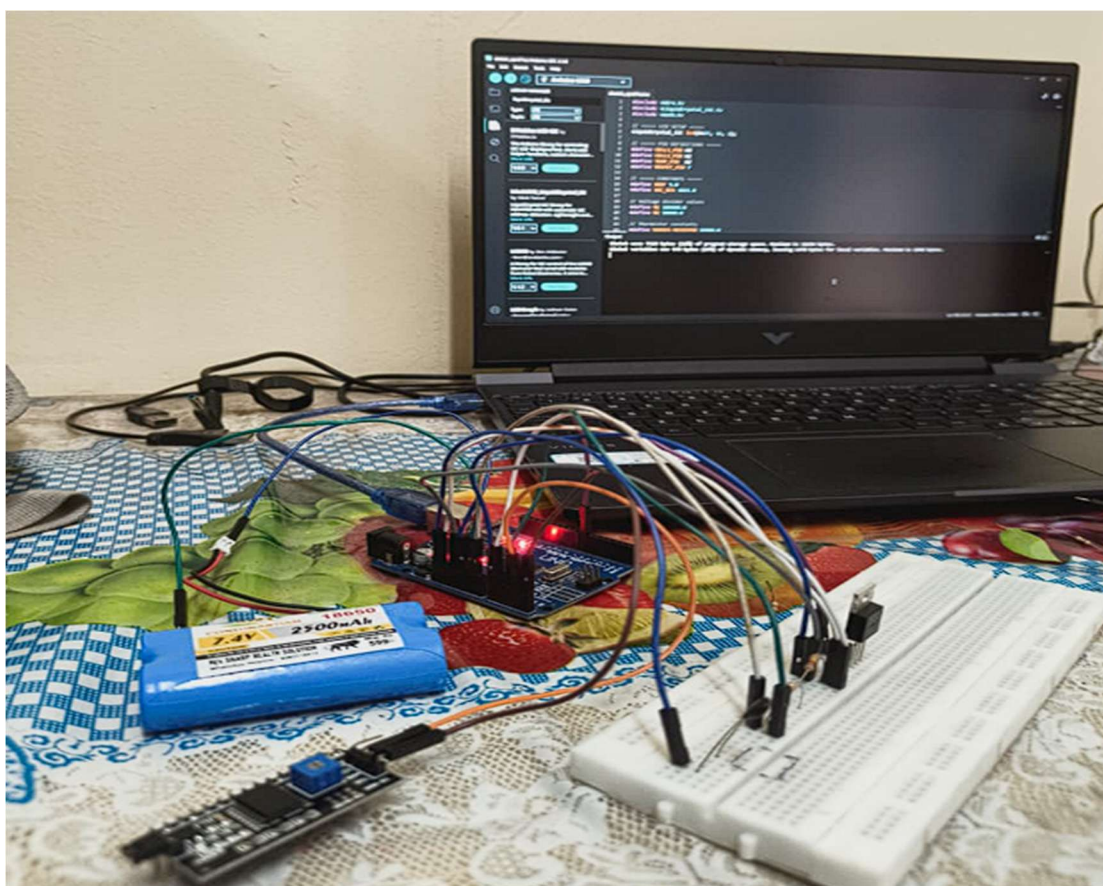
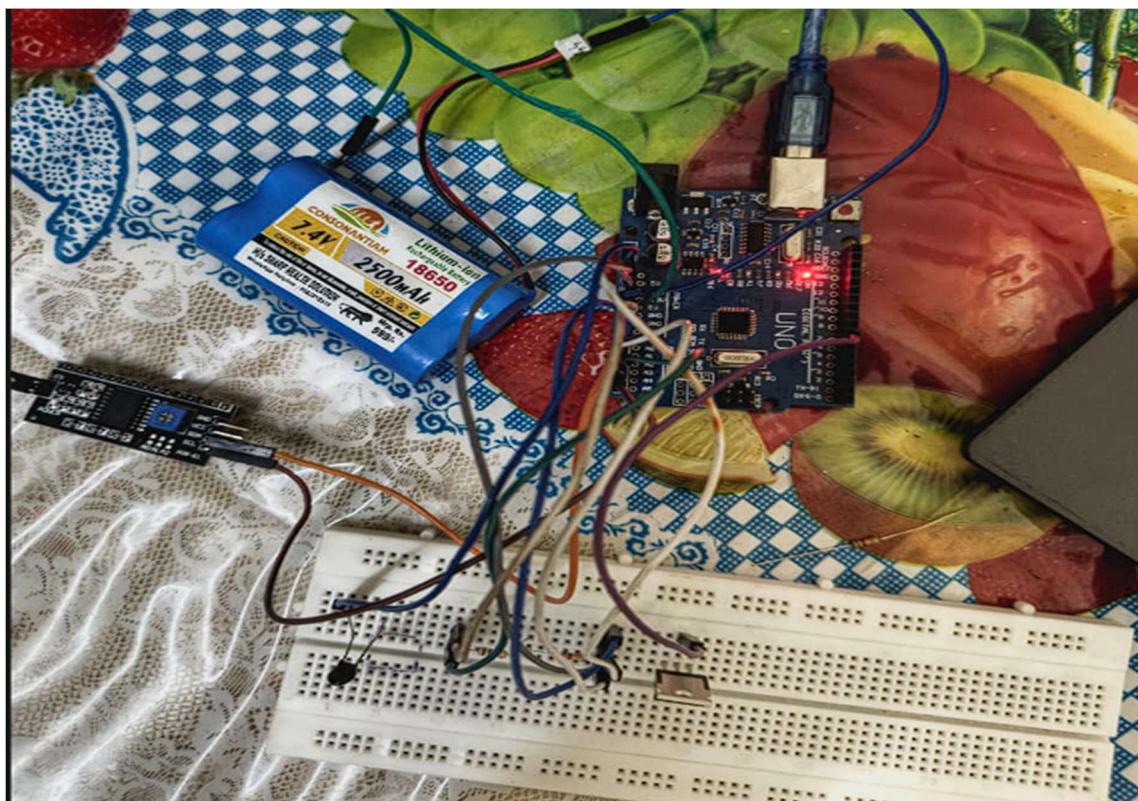
lcd.print(" "); // clear extra chars

```

```
lcd.setCursor(0, 1);  
  
lcd.print("T:");  
  
lcd.print(temperature, 1);  
  
if (fault) {  
    lcd.print(" FAULT ");  
}  
else {  
    lcd.print(" OK  ");  
}  
  
delay(1000);  
}
```

Output Observations

- Real-time display of battery parameters was achieved
- Accurate detection of abnormal conditions
- Fast response of MOSFET switching
- Stable system performance during testing



Conclusion of Testing

The implemented BMS successfully monitored and controlled battery parameters. The system responded effectively to unsafe conditions and provided reliable protection, demonstrating the feasibility of a low-cost Battery Management System.

CHAPTER 5: RESULTS & DISCUSSION

5.1 Result Analysis

The developed Battery Management System (BMS) was tested under various operating conditions to evaluate its performance in monitoring and protecting the battery. The system successfully measured key parameters such as voltage, current, and temperature in real-time and displayed them on the LCD.

Voltage Measurement

The voltage divider circuit enabled accurate scaling of battery voltage within the measurable range of the Arduino. The observed values were consistent with expected readings, with minimal deviation.

Current Measurement

The shunt resistor effectively measured current flowing through the load. The system was able to detect variations in current and respond appropriately when the threshold limit was exceeded.

Temperature Monitoring

The NTC thermistor showed a clear inverse relationship between temperature and resistance. As temperature increased, the system accurately detected the change and triggered protection when the threshold was crossed.

Protection Mechanism

The MOSFET switching mechanism worked efficiently:

- Under normal conditions → MOSFET remained ON
- Under fault conditions (overtemperature, overcurrent, overvoltage) → MOSFET turned OFF

The system demonstrated a quick response time, ensuring safe operation of the battery.

Graphical Interpretation (to draw in report)

- Voltage vs Time
- Current vs Time
- Temperature vs Time

These graphs show stable operation under normal conditions and sharp changes when thresholds are exceeded.

5.2 Performance Evaluation

The performance of the proposed BMS was compared with existing systems based on key parameters such as cost, complexity, accuracy, and functionality.

Discussion

The proposed system provides a balanced solution between cost and performance. Unlike basic analog systems, it offers real-time monitoring and better accuracy due to the use of a microcontroller. At the same time, it avoids the complexity and high cost associated with smart BMS systems.

While advanced BMS solutions include features such as wireless communication and cloud integration, they are not always necessary for small-scale applications. The proposed system fulfills essential safety requirements while remaining simple and affordable.

The system performs efficiently for educational purposes and low-power applications. However, it can be further improved by incorporating advanced features such as cell balancing, data logging, and wireless monitoring.

Overall Observation

The results confirm that the designed BMS is:

- Reliable for monitoring key battery parameters
- Effective in protecting against unsafe conditions
- Cost-efficient and suitable for small-scale applications

This demonstrates that a simple and well-designed system can provide significant improvements in battery safety and performance.

CHAPTER 6: CONCLUSION & FUTURE SCOPE

6.1 Summary of Work and Achievements

This project presented the design and implementation of a **Battery Management System (BMS)** using an NTC thermistor, shunt resistor, MOSFET, and Arduino-based control system. The system was developed to monitor critical battery parameters such as voltage, current, and temperature, and to provide protection against unsafe operating conditions.

The implemented system successfully:

- Measured real-time battery parameters
- Displayed values using an LCD interface
- Detected abnormal conditions such as overtemperature, overcurrent, and overvoltage
- Performed automatic cutoff using a MOSFET to protect the battery

The project demonstrates a cost-effective and efficient solution suitable for small-scale applications and educational purposes.

6.2 Impact on Society, Environmental Sustainability, Ethical Issues and Compliance

The proposed BMS contributes positively to both society and the environment. From a societal perspective, it enhances safety by reducing the risks associated with battery failures such as overheating and short circuits. This is particularly important in applications like portable electronics and energy storage systems.

From an environmental standpoint, efficient battery management increases battery lifespan, thereby reducing electronic waste and minimizing environmental pollution

caused by battery disposal. It also supports the adoption of renewable energy systems by ensuring safe energy storage.

Ethically, the system promotes responsible use of battery technology by preventing hazardous conditions. The project aligns with industry practices by focusing on safety, reliability, and efficiency. Although it is a basic implementation, it follows fundamental safety standards relevant to battery-operated systems.

6.3 Limitations of the Work

Despite its effectiveness, the project has certain limitations:

- It is designed for a single battery system and does not support multi-cell battery packs
- Advanced features such as active cell balancing are not implemented
- SOC and SOH estimation are basic and not highly accurate
- The system does not include wireless monitoring or remote access
- Accuracy depends on sensor calibration and may vary slightly
- Limited scalability for high-power industrial applications

6.4 Future Scope

The proposed system can be further enhanced in several ways:

- Implementation of advanced SOC and SOH estimation algorithms for improved accuracy
- Integration of active cell balancing techniques for multi-cell battery systems
- Addition of wireless communication (Bluetooth/Wi-Fi) for remote monitoring

- Development of a mobile or web-based interface for real-time data visualization
- Incorporation of data logging for performance analysis and predictive maintenance
- Use of more precise sensors and dedicated BMS ICs for improved reliability
- Expansion of the system for electric vehicle and large-scale energy storage applications

Final Conclusion

The project successfully demonstrates that a simple, low-cost Battery Management System can effectively monitor and protect batteries. It provides a strong foundation for further development into more advanced and intelligent BMS solutions, contributing to safer and more sustainable use of battery technology.

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